

Solution Requirements (Functional & Non-functional)

Date	29/09/2026
Team ID	-
Project Name	BlogNest - AI-Powered Blog Management Backend
Maximum Marks	-

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Register with name, email and password; passwords hashed with bcrypt.js; default role is reader
FR-2	User Authentication	Login with email and password; JWT issued and validated on protected routes
FR-3	Role-Based Access Control	Admin, Editor, Author and Reader permissions enforced by middleware; admin can change roles
FR-4	Blog Management	Create, read, update, delete posts; manage lifecycle status (Draft, Pending Approval, Scheduled, Published); attach media path
FR-5	Categories and Tags	Assign categories and tags to posts; filter and search posts by them
FR-6	Comment Management	Readers add comments; editors and admins moderate (approved, pending, spam) and prune malicious ones
FR-7	AI Blog Generation	Generate a structured blog from a theme via /api/ai/generate-blog using Google Gemini
FR-8	AI Summarization	Summarize long articles into concise abstracts via /api/ai/summarize
FR-9	Post Performance Tracking	Track likes and engagement data so authors can review post performance

Non-functional Requirements:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Consistent, predictable JSON responses and clear error messages; easy to test with Postman and to consume from any frontend.
NFR-2	Security	JWT authentication, role-based authorization, bcrypt password hashing, input sanitization against XSS and NoSQL injection, request validation, rate limiting.
NFR-3	Reliability	Centralized error handling, schema validation through Mongoose, and structured logging to detect and recover from failures.
NFR-4	Performance	Lightweight Express routing, indexed MongoDB queries, and offloading AI text processing to an external service.
NFR-5	Availability	Stateless server design allows multiple instances behind a load balancer; MongoDB can be deployed with replication.
NFR-6	Scalability	Modular MVC architecture that scales horizontally, with new services and routes added independently.